Conception Phase- Airbnb Booking Platform

By Mohamed Amine Boughattas

Matriculation: 10244528

This project aims to design a relational data model for an Airbnb-style hotel booking system. The concept centers on simulating a real-world online accommodation platform where hosts can list properties and guests can search, book, and review stays. The primary objective of this phase is to define the functional scope, entities, and data relationships of the system. The final deliverable of this phase includes a comprehensive Entity Relationship Model (ERM), an abstract, and the foundation for future implementation in SQL.

Motivation and Goal

The motivation behind this database model is to capture the complexity of a real-life digital accommodation marketplace while enabling data analytics, traceability, and business process automation. The key goal is to build a robust, normalized database structure that supports multiple user roles, booking flows, review systems, and financial tracking providing a realistic foundation for data warehousing, mart creation, and later analytics.

Core Concept and Methodology

The proposed model includes **21 well-connected entities**, encompassing all operational aspects of the Airbnb system. A top-down design methodology is used beginning with a functional decomposition of Airbnb's core services into entities and relationships. From there, we mapped each user interaction (e.g., host creating a listing, guest booking, user reviewing) into data structures and constraints. It followed **IU's ER modeling expectations** closely, ensuring that the model includes:

- At least **20 entities**
- Multiple **triple relationships** (e.g., reviews: guest, property, host)
- A **recursive relationship** (users referring other users)
- Clearly defined **foreign key constraints**
- Normalization to at least **3NF** to prevent data redundancy

The tool used is **dbdiagram.io**, which supports clean visual modeling and DBML-based SQL generation. The resulting diagram (see attached ERD) maps user roles, data flows, and relationships clearly. Composite keys are defined using the correct syntax (Indexes block), and all attributes are labeled with appropriate types. The model supports forward scalability into data marts or warehouse dimensions.

System Overview

Users of the system include:

- **Hosts**, who list and manage properties
- **Guests**, who browse, book, and review stays
- **Admins**, who may handle escalations and system data

Key data actions and flows:

- Guests can create **bookings** for properties
- Both guests and hosts can write **reviews**
- Payments are processed per booking and recorded along with **platform commission**
- Listings include **rules**, **photos**, **availability calendars**, and **amenities**
- Users can send **messages**, add listings to **wishlists**, and report **complaints**
- Host performance is tracked through **income reports** and **ID verification**
- A **referral** system is built-in, where users can invite others to join

The relationships and business logic are reflected in the schema. For example:

- **Bookings** link guests and properties with check-in/out dates and status
- **Reviews** involve a triple relation among guest, host, and property
- **Wishlist items** and **property amenities** demonstrate many-to-many logic
- **Recursive relationship** in users.referred_by tracks referral chains

This conception phase concludes with a full UML-style ERD, designed in dbdiagram.io and compliant with IU's portfolio guidelines.

The deliverables include a brief concept text, the ER model diagram, and a database dictionary

All constraints, relationships, and data flows are thoroughly mapped

The next step will beginning the development and reflection phase. This early groundwork ensures the database will support analytical use cases such as host performance tracking, booking trends, and user engagement insights.

Data Dictionary:

1. users

Attribute	Type	Key	Description
id	int	PK	Unique identifier for each platform user
name	varchar(100)		Full name of the user
email	varchar(255)	UQ	Unique email address used for login/contact
password	varchar(255)		Encrypted or stored password value
phone	varchar(30)		Contact phone number
created_at	datetime		Date and time the user account was created
referred_by	int	FK	Self-reference to the user who referred this user

2. hosts

Attribute	Type	Key	Description
user_id	int	PK, FK	Host identifier linked to users.id
host_since	date		Date when the user started hosting
overall_rating	decimal(3,2)		Average rating received as a host
is_superhost	boolean		Indicates whether the host has superhost status

3. guests

Attribute	Type	Key	Description
user_id	int	PK, FK	Guest identifier linked to users.id
preferred_language	varchar(50)		Preferred language of the guest
default_currency	varchar(10)		Main currency used by the guest

4. properties

Attribute	Type	Key	Description

id	int	PK	Unique property identifier
host_id	int	FK	Host who owns or manages the property
title	varchar(255)		Property listing title
description	text		Detailed textual description
type	varchar(50)		Type of property, such as apartment or house
address	text		Full street address
city	varchar(100)		City where the property is located
country	varchar(100)		Country where the property is located
price_per_night	decimal(10,2)		Price per night for booking
created_at	datetime		Listing creation timestamp

5. amenities

Attribute	Type	Key	Description
id	int	PK	Unique amenity identifier
name	varchar(100)	UQ	Amenity name, e.g. WiFi or Parking

6. property_amenities

Attribute	Type	Key	Description
id	int	PK	Unique record identifier
property_id	int	FK	Property that offers the amenity
amenity_id	int	FK	Amenity assigned to the property

7. calendars

Attribute	Type	Key	Description
id	int	PK	Unique calendar row identifier
property_id	int	FK	Property linked to the availability record

available_date	date		Specific calendar date
is_available	boolean		Availability status for that date

8. bookings

Attribute	Type	Key	Description
id	int	PK	Unique booking identifier
guest_id	int	FK	Guest who created the booking
property_id	int	FK	Booked property
checkin_date	date		Planned check-in date
checkout_date	date		Planned check-out date
total_price	decimal(10,2)		Total booking cost
status	varchar(50)		Booking state, e.g. completed or cancelled
created_at	datetime		Timestamp of booking creation

9. payments

Attribute	Type	Key	Description
id	int	PK	Unique payment identifier
booking_id	int	FK	Booking linked to the payment
amount	decimal(10,2)		Payment amount paid through platform
payment_method	varchar(50)		Payment type, e.g. card or PayPal
paid_at	datetime		Payment timestamp

10. platform_commissions

Attribute	Type	Key	Description
id	int	PK	Unique commission identifier
booking_id	int	FK	Booking tied to commission charges

guest_fee	decimal(10,2)		Fee charged to guest
host_fee	decimal(10,2)		Fee charged to host

11. incomes

Attribute	Type	Key	Description
id	int	PK	Unique income record identifier
host_id	int	FK	Host receiving the income
month	varchar(20)		Month of the income record
year	int		Year of the income record
income	decimal(10,2)		Net or summarized host income

12. reviews

Attribute	Type	Key	Description
id	int	PK	Unique review identifier
reviewer_id	int	FK	User who writes the review
reviewee_id	int	FK	User being reviewed
property_id	int	FK	Property related to the review
rating	int		Numeric rating value
comment	text		Written feedback text
created_at	datetime		Timestamp of review submission

13. favorites

Attribute	Type	Key	Description
id	int	PK	Unique favorite record identifier
guest_id	int	FK	Guest who marked the property as favorite
property_id	int	FK	Favorited property

favorited_at	datetime		Date and time property was favorited

14. wishlists

Attribute	Type	Key	Description
id	int	PK	Unique wishlist identifier
guest_id	int	FK	Guest who owns the wishlist
name	varchar(200)		Wishlist title or label
created_at	datetime		Wishlist creation timestamp

15. wishlist_items

Attribute	Type	Key	Description
id	int	PK	Unique wishlist item identifier
wishlist_id	int	FK	Wishlist containing the saved property
property_id	int	FK	Property saved into the wishlist

16. messages

Attribute	Type	Key	Description
id	int	PK	Unique message identifier
sender_id	int	FK	User who sent the message
receiver_id	int	FK	User who received the message
message	text		Message body text
sent_at	datetime		Sending timestamp

17. complaints

Attribute	Type	Key	Description
id	int	PK	Unique complaint identifier
booking_id	int	FK	Booking related to the complaint

user_id	int	FK	User who raised the complaint
complaint_text	text		Complaint description
resolved	boolean		Resolution status

18. host_verifications

Attribute	Type	Key	Description
id	int	PK	Unique verification record identifier
host_id	int	FK	Host being verified
id_document_url	varchar(500)		Path or URL to identity document
verified	boolean		Indicates verification result
verified_at	datetime		Timestamp of verification

19. social_profiles

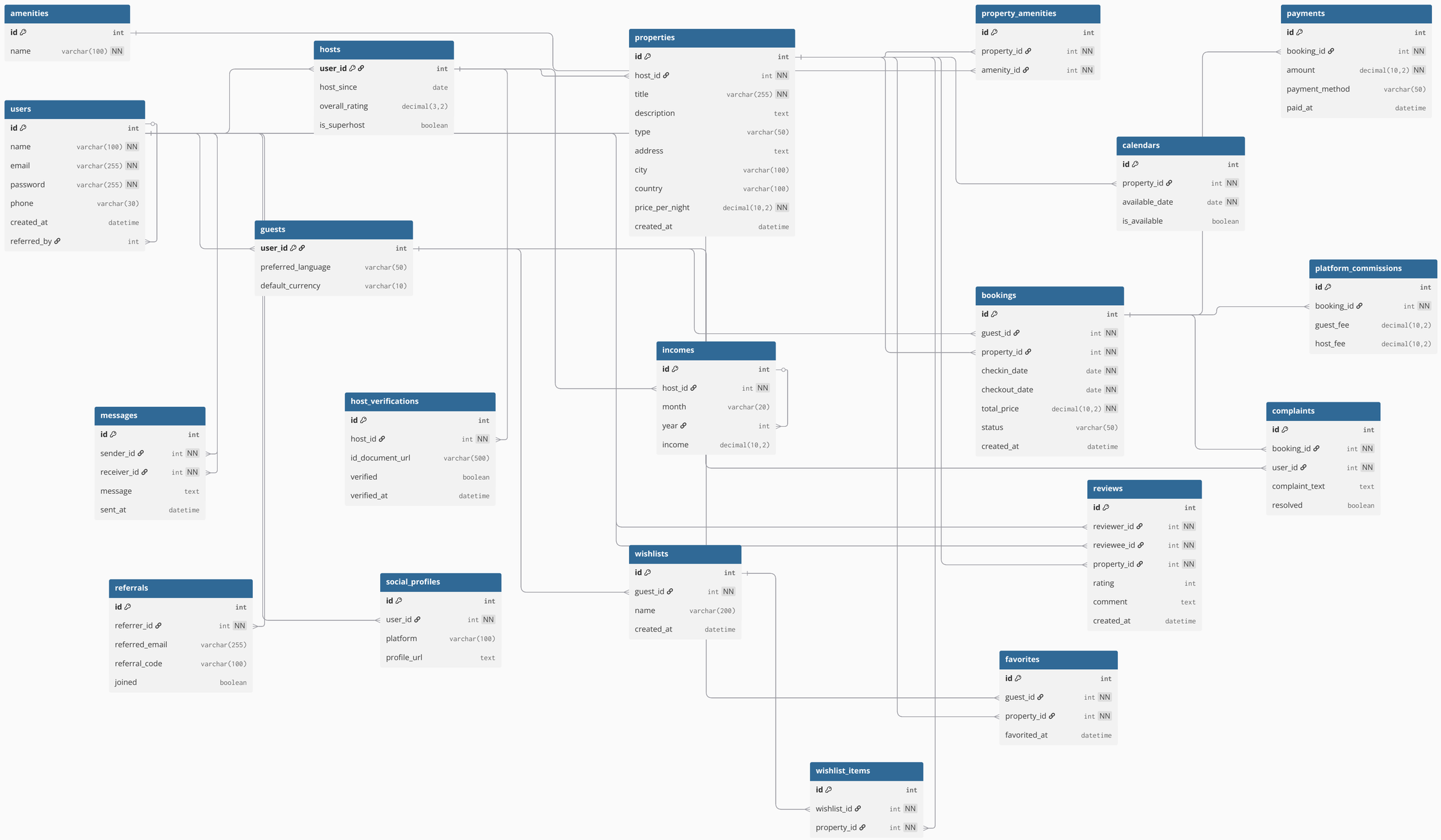
Attribute	Type	Key	Description
id	int	PK	Unique social profile identifier
user_id	int	FK	User linked to the social account
platform	varchar(100)		Social platform name
profile_url	text		URL to the connected social profile

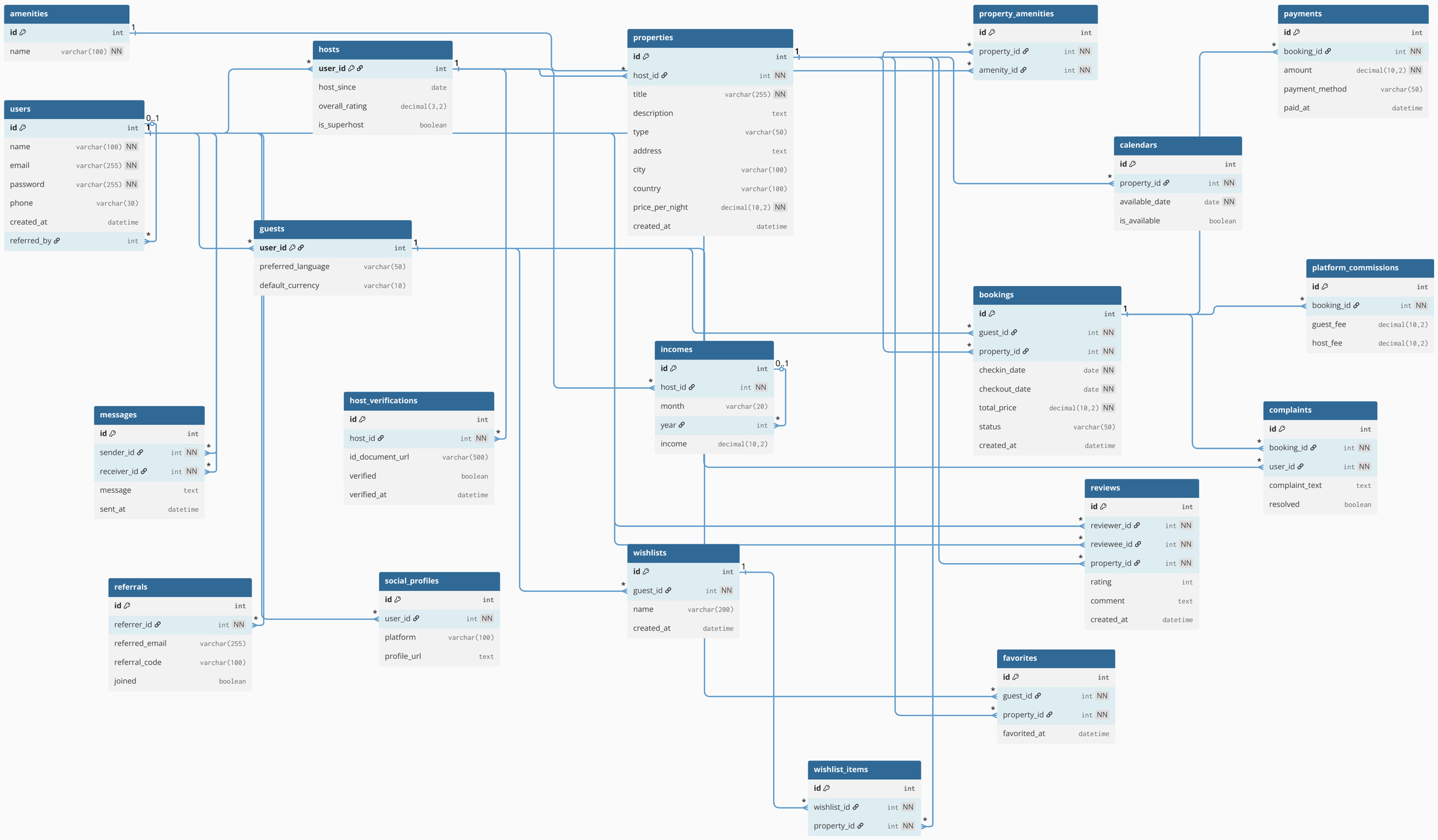
20. referrals

Attribute	Type	Key	Description
id	int	PK	Unique referral record identifier
referrer_id	int	FK	User who invited another user
referred_email	varchar(255)		Email of referred person
referral_code	varchar(100)		Generated referral code
joined	boolean		Indicates whether referred person joined

The original conception correctly identified the main Airbnb business processes, such as user registration, hosting, booking, payment handling, reviews, messaging, wishlists, complaints, and referral logic. The model also already covered many-to-many structures like property amenities and wishlist items, as well as recursive referral behavior in the users entity. However, the initial role design assumed that one user could only be one type of participant at a time. Based on tutor feedback, this was refined using a joined subclass inheritance strategy. A general `users` entity now stores shared profile data, while `hosts` and `guests` act as subtype entities linked to `users`. This correction makes the ER model more realistic, because a person on Airbnb can act both as a guest and as a host. The refined model remains normalized, preserves the original business logic, and better reflects real platform behavior.

Updated ERM


amenities
id int
name varchar(100) NN
hosts
user_id int
host_since date
overall_rating decimal(3,2)
is_superhost boolean
properties
id int
host_id int NN
title varchar(255) NN
description text
type varchar(50)
address text
city varchar(100)
country varchar(100)
price_per_night decimal(10,2) NN
created_at datetime
property_amenities
id int
property_id int NN
amenity_id int NN
payments
id int
booking_id int NN
amount decimal(10,2) NN
payment_method varchar(50)
paid_at datetime
users
id int
name varchar(100) NN
email varchar(255) NN
password varchar(255) NN
phone varchar(30)
created_at datetime
referred_by int
calendars
id int
property_id int NN
available_date date NN
is_available boolean
guests
user_id int
preferred_language varchar(50)
default_currency varchar(10)
platform_commissions
id int
booking_id int NN
guest_fee decimal(10,2)
host_fee decimal(10,2)
bookings
id int
guest_id int NN
property_id int NN
checkin_date date NN
checkout_date date NN
total_price decimal(10,2) NN
status varchar(50)
created_at datetime
incomes
id int
host_id int NN
month varchar(20)
year int
income decimal(10,2)
host_verifications
id int
host_id int NN
id_document_url varchar(500)
verified boolean
verified_at datetime
messages
id int
sender_id int NN
receiver_id int NN
message text
sent_at datetime
complaints
id int
booking_id int NN
user_id int NN
complaint_text text
resolved boolean
reviews
id int
reviewer_id int NN
reviewee_id int NN
property_id int NN
rating int
comment text
created_at datetime
wishlists
id int
guest_id int NN
name varchar(200)
created_at datetime
referrals
id int
referrer_id int NN
referred_email varchar(255)
referral_code varchar(100)
joined boolean
social_profiles
id int
user_id int NN
platform varchar(100)
profile_url text
favorites
id int
guest_id int NN
property_id int NN
favorited_at datetime
wishlist_items
id int
wishlist_id int NN
property_id int NN
dbdiagram.io


amenities
id int
name varchar(100) NN
hosts
user_id int
host_since date
overall_rating decimal(3,2)
is_superhost boolean
properties
id int
host_id int NN
title varchar(255) NN
description text
type varchar(50)
address text
city varchar(100)
country varchar(100)
price_per_night decimal(10,2) NN
created_at datetime
property_amenities
id int
property_id int NN
amenity_id int NN
payments
id int
booking_id int NN
amount decimal(10,2) NN
payment_method varchar(50)
paid_at datetime
users
id int
name varchar(100) NN
email varchar(255) NN
password varchar(255) NN
phone varchar(30)
created_at datetime
referred_by int
calendars
id int
property_id int NN
available_date date NN
is_available boolean
guests
user_id int
preferred_language varchar(50)
default_currency varchar(10)
platform_commissions
id int
booking_id int NN
guest_fee decimal(10,2)
host_fee decimal(10,2)
bookings
id int
guest_id int NN
property_id int NN
checkin_date date NN
checkout_date date NN
total_price decimal(10,2) NN
status varchar(50)
created_at datetime
incomes
id int
host_id int NN
month varchar(20)
year int
income decimal(10,2)
messages
id int
sender_id int NN
receiver_id int NN
message text
sent_at datetime
host_verifications
id int
host_id int NN
id_document_url varchar(500)
verified boolean
verified_at datetime
complaints
id int
booking_id int NN
user_id int NN
complaint_text text
resolved boolean
reviews
id int
reviewer_id int NN
reviewee_id int NN
property_id int NN
rating int
comment text
created_at datetime
wishlists
id int
guest_id int NN
name varchar(200)
created_at datetime
referrals
id int
referrer_id int NN
referred_email varchar(255)
referral_code varchar(100)
joined boolean
social_profiles
id int
user_id int NN
platform varchar(100)
profile_url text
favorites
id int
guest_id int NN
property_id int NN
favorited_at datetime
wishlist_items
id int
wishlist_id int NN
property_id int NN
dbdiagram.io